🔑 **Answer Key — Teacher Copy · Lesson 3-5**

MSTAR PRACTICE WORKSHEET

Lesson 3-5: Compare Ratio Relationships

Grade 6 Mathematics · Standard 6.AT.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?

A. Chef A

B. Chef B

C. They use the same amount

D. Cannot determine

Chef A: 2÷8 = 0.25 cups per cracker. Chef B: 3÷9 ≈ 0.33 cups per cracker. Chef B uses more cheese per cracker.

Question 2 SELECTED RESPONSE — CORRECT: B

Which ratio is greater: 3:4 or 5:8?

A. 5:8

B. 3:4

C. They are equal

D. Cannot determine

Convert to the same denominator: 3:4 = 6:8. Since 6:8 > 5:8, the ratio 3:4 is greater.

Question 3 SELECTED RESPONSE — CORRECT: A

Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.

- (A) Recipe B
- (B) Recipe A
- (C) They are the same
- (D) Cannot determine

Recipe A: 2÷6 ≈ 0.33 lemons per cup. Recipe B: 3÷7 ≈ 0.43 lemons per cup. Since 0.43 > 0.33, Recipe B is more lemony.

Question 4 SELECTED RESPONSE — CORRECT: B

Which is the better deal: 4 apples for $3 or 6 apples for $5?

- (A) 6 apples for $5
- (B) 4 apples for $3
- (C) They cost the same
- (D) Cannot determine

Deal 1: $3÷4 = $0.75 per apple. Deal 2: $5÷6 ≈ $0.83 per apple. $0.75 < $0.83, so 4 apples for $3 is the better deal.

Question 5 SELECTED RESPONSE — CORRECT: C

Store A sells 3 mangoes for $6. Store B sells 5 mangoes for $9. Which store offers a better price per mango?

- (A) Store A
- (B) Same price
- (C) Store B
- (D) Cannot determine

Store A: $6÷3 = $2.00 per mango. Store B: $9÷5 = $1.80 per mango. Store B has the better (lower) price per mango.

Question 6 SELECTED RESPONSE — CORRECT: B

Chef Academy's snack bar sells 3 mini quiches for $6 or 5 mini quiches for $8. Which is the better deal per quiche?

A 3 quiches for $6

B 5 quiches for $8

C They cost the same

D Cannot determine

3 for $6 is $6 / 3 = $2.00 per quiche. 5 for $8 is $8 / 5 = $1.60 per quiche. $1.60 < $2.00, so 5 quiches for $8 is the better deal.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

Car A travels 180 miles on 6 gallons. Car B travels 225 miles on 9 gallons. Which car gets better gas mileage?

A. Car B, at 25 miles per gallon

B. Car B, because it traveled farther

C. They are equal

D. Car A, at 30 miles per gallon

Car A: 180 ÷ 6 = 30 miles per gallon. Car B: 225 ÷ 9 = 25 miles per gallon. 30 > 25, so Car A gets better mileage.

- **A:** 25 miles per gallon is Car B's rate and it is computed correctly, but 30 is the larger rate.
- **B:** Car B did travel farther, but it burned more gas doing it. Compare miles PER GALLON.
- **C:** 180 divided by 6 is 30 and 225 divided by 9 is 25, so the two rates differ.

Part B — correct answer: C

Why is comparing total miles alone NOT enough to decide which car is more efficient?

A. Total miles is never a useful measurement.

B. The car with fewer miles is always more efficient.

C. The cars used different amounts of gas, so the totals are not comparable until each is expressed per gallon.

D. Efficiency depends only on the size of the gas tank.

Car B traveled farther but also used more gas. Converting each to a unit rate puts both on the same per-gallon basis, which is what makes them comparable.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Comparing Totals Instead of Rates

> Jaden compared two paint sizes: a 3-gallon can for $54 and a 5-gallon can for $85. He wrote: 'The 3-gallon can is the better deal because $54 is less than $85.'

Explain Jaden's misconception and determine which can is actually the better deal.

Model answer: 54 ÷ 3 = $18 per gallon; 85 ÷ 5 = $17 per gallon. The 5-gallon can is the better deal. Jaden compared totals for different amounts of paint; only the price per gallon compares them fairly.

2 points	Student computes both unit prices ($18 per gallon and $17 per gallon), identifies the 5-gallon can as the better deal, and explains that a lower total price does not mean a lower price per gallon.
1 point	Student identifies the 5-gallon can but does not show or explain the unit prices.
0 points	Student agrees with Jaden or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Store A sells 5 notebooks for $12. Store B sells 8 notebooks for $20. Without calculating, a student says Store B must be cheaper because you get more notebooks. Is the student's reasoning valid? Find the unit rates to prove your answer.

Model answer: The student's reasoning is not valid — getting more items does not mean a lower price per item. Store A: $12 ÷ 5 = $2.40 per notebook. Store B: $20 ÷ 8 = $2.50 per notebook. Store A is actually cheaper per notebook, even though Store B offers more notebooks per purchase.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.